A Building Is Not Its Silhouette: Structural Blindness in 3D Generation Evaluation

Anonymous CVPR submission

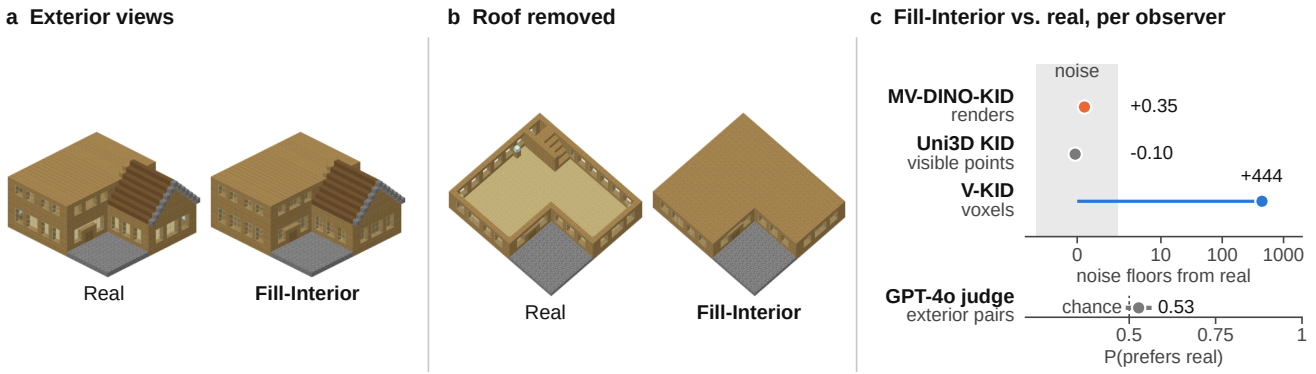


Figure 1. **A metric ranks only what it observes. (a)** A real held-out house and its FILL-INTERIOR copy, in which every sealed room is filled with the house's own material (+138% blocks on this build; median +33% over 200 held-out houses). Rendered in flat colours, the four metric views of this build are bit-identical to those of its copy. **(b)** With the roof removed, the real house has rooms and the copy is solid. **(c)** Distance of the filled corpus from real data under each observer, in noise floors (the shaded band is ±2). The multi-view DINOv2 KID on exterior renders and Uni3D on visible-surface points (n=128; seeds −0.1 and +0.4) stay inside the noise, and a GPT-4o judge shown exterior pairs prefers the real house in 0.53 of trials (95% CI [0.50, 0.56]). V-KID on voxels places the same corpus 444 noise floors away. The render and voxel values come from the n=256 leaderboard (§6.4); published builds come from 3D-Craft (Supp. Sec. M).

Abstract

Generative models of 3D structure are ranked by metrics that observe only part of what they score: rendered exterior views, points sampled from visible surfaces, or a judge shown images. A metric cannot rank what its observations discard, and for buildings, LEGO models and indoor scenes the discarded part is often the interior, which decides whether a structure can be entered, assembled or navigated. We make this limit measurable with complement probes that alter a structure while keeping every exterior-visible face identical. Filling the sealed rooms of real Minecraft houses adds a median 33% more blocks, yet a multi-view DINOv2 kernel distance places the filled corpus 0.35 noise floors (units of the metric's own real-versus-real sampling spread) from real data, in the same significance group. The blindness holds for seven render statistics, five image encoders and up to sixteen views, for Uni3D and ULIP-2 point-cloud encoders and Chamfer and earth mover's distances on visible-surface points, and for a GPT-4o judge, across Minecraft, LEGO and Objaverse meshes. Changing only the observation, by cutting the views open or sampling interior faces, recovers the signal. We introduce V-KID, a CPU-only voxel metric that separates the probe by 444 noise floors and is blind exactly where render metrics see, and a protocol that admits a metric to a leaderboard only after it passes gates on ordering, invariance and sample-size stability, which Fréchet distance fails. Applied to the published generator BrickGPT, the protocol finds a geometric gap 2.2 times its render gap in effect size.

1. Introduction

Many of us are taught at a young age that we shouldn't judge a book by its cover. A building can be examined the same way. It may have rooms, doors–an interior. And, in many cases, structures rely on these interior layouts to make them usable rather than decorative. Consider a Minecraft build someone walks through, a LEGO model someone assembles, or an indoor environment that needs navigable space and collision volumes for robotic training.

Generative models of discrete 3D structure have advanced quickly, over LEGO assemblies [18, 28, 33], sparse voxel categories [17], Minecraft worlds [22] and serialized geometry [30, 31]. Many of their evaluation techniques come from image generation where they render a structure from a handful of views, embed the renders with a pretrained encoder such as DINOv2 [26], and report a Fréchet inception distance (FID) [14] or kernel inception distance (KID) [4] between the generated and real feature distributions. We call this family *render-and-embed*. Its limitation is already on the record. The largest generative study in this domain [22] lists among its own limitations that render-based FID "struggles to assess internal structure (e.g., whether a building has a coherent interior)," then ranks models with it for want of an alternative. What the field lacks is a measurement of how large the effect is and a metric that sees what it misses. We supply both.

Our probe, FILL-INTERIOR, takes real buildings and fills every enclosed air cell with that build's own dominant material, so no new material betrays the corruption (Figure 1). It adds a median of 33% more blocks, alters 94% of examined builds, and leaves the exterior silhouette untouched. Scored at n=256 against 399 held-out real builds, a solidified corpus and its genuine, real data counterpart *share a significance group* under a paired, Comment: What does this mean haha? These two last lines Holm-corrected [15] comparison over all arm pairs. The same encoder and the same statistic separate a material permutation of the same builds at +422.43 noise floors (Table 4).

Comment: weakish? Hard to follow–maybe too much bloat, see what really adds something in this paragraph The failure is in what the views show, not in what the encoder computes. Holding the encoder and the statistic fixed and changing only the cameras, so that the renders expose the interior, separates the same probe at 6.7 to 55.8 noise floors (§6.2). Under an opaque renderer the invisibility is exact as filling sealed voids cannot change any exterior view, so no encoder, statistic, sample size or view budget separates a corpus from its solidified copy (§6.2, Supp. Sec. H).

Comment: what does this line mean A metric should therefore earn the right to rank. We score candidate metrics against corruptions of known severity and require monotonicity, invariance and sample-size stability before any of them orders a scoreboard, with separate disqualification gates for memorization and mode collapse. Measuring geometry directly, alongside the renders rather than instead of them, closes the blind spot. We make three main contributions.

1. **Interior blindness, and it belongs to the observation** (§6.1–6.3). Render-and-embed distances cannot distinguish a corpus of real buildings from the same corpus with every interior filled or removed, while the same test resolves every corruption that reaches the exterior. Seven render-space statistics, five encoders and view budgets up to sixteen views fail the same probes, and so do Uni3D, ULIP-2, chamfer and EMD on visible-surface points and a GPT-4o judge, on Minecraft, LEGO and Objaverse. Cutaway renders and interior-face sampling recover the signal with the same encoder.
2. **A geometric metric with the complementary blindness** (§4). V-KID is a kernel distance on D4-invariant voxel descriptors. It separates the probe by 444 noise floors, needs no GPU, and is *exactly* invariant to material permutation, so neither metric alone ranks both corruptions correctly.
3. **A validation protocol, and a suite that implements it** (§4). Gates on ordering, invariance and n-stability decide whether a metric may rank anything at all. Fréchet distance, the functional form FID uses, fails n-stability, as do coverage and the biased maximum mean discrepancy (MMD) estimator [16]; precision, density and recall [19, 23] fail monotonicity.

2. Related Work

Comment: review the flow here Recent systems for generating discrete 3D structures, including text-to-brick systems [18, 24, 28], voxel generators [2, 8, 10, 17], and Minecraft world models [10], typically evaluate their outputs through rendered views.

Render-based evaluation of discrete 3D generation. Dream-Cubed [22], for example, evaluates generated Minecraft worlds using render-based FID, and many text-to-brick systems use rendered or structural representations to assess generated assemblies [18, 24, 28]. FID [14] and KID [4] are the default two-sample statistics. Notably, Dream-Cubed also names the limitation we measure in its own limitations section and reports raters agreeing with that FID only 54.3% of the time. At our sample sizes the unbiased MMD estimator [12] is preferable, and §4 shows that a widely copied biased variant ranks by sample size instead. Precision and recall [19], density and coverage [23] and CMMD [16] vary the statistic and the features but stay in image space, so all inherit the blindness we measure.

Alternatives to image-space evaluation. A natural alternative to rendering is to evaluate generated geometry directly. Point-cloud methods compare shape distributions in geometric space [1], avoiding some of the information loss introduced by projection to images. However, changing the representation does not by itself guarantee sensitivity to the property of interest. Luan et al. [20] similarly identify limitations of render-based evaluation and propose surface-point sampling to improve view coverage. Their representation remains restricted to the observed boundary:

in our setting, a structure and its interior-filled counterpart have identical exterior surfaces, and therefore receive identical surface-point representations. Thus, this approach addresses incomplete *view coverage* but not blindness to geometry that does not alter the exterior surface. Learned point-cloud encoders such as Uni3D [35] and ULIP-2 [34] inherit the same limit through their sampler, which we measure directly (§6.1). Our geometric evaluation instead operates on volumetric structure, allowing it to detect the interior changes introduced by our controlled corruption.

Auditing generative-model metrics. Several works have argued that generative-model metrics should be tested for desirable properties rather than treated as universally valid measures of quality. Borji [5] identifies sensitivity to meaningful distortions and invariance to irrelevant transformations as desirable metric properties, while Chong and Forsyth [9] analyze the dependence of FID on sample size. O'Bray et al. [25] provide a closer methodological precedent by applying controlled perturbations of increasing severity and testing whether a metric responds monotonically. They use these, however, to select metric hyperparameters where we, instead, use controlled structural corruptions as a precondition for evaluation. We examine how a metric must demonstrate sensitivity to relevant structural changes and invariance to irrelevant ones before being used to compare generators. Hervé and Salge [13] provide a closer in-domain example of metric evaluation for procedural content generation, while Caytuiro and Sipiran [7] audit whether generated structures exhibit a particular structural property. Our focus differs in that we audit the *observation and representation pipeline itself*: specifically, whether an evaluation protocol can detect changes to 3D geometry that are invisible in its input views

Memorization and human evaluation. Other work addresses complementary problems that arise when evaluating generative systems. Memorization can produce deceptively strong realism scores [3, 6], motivating nearest-neighbor retrieval and held-out comparisons for 3D generation [21, 27]. Human judgments provide another form of validation; GDMC [29], for example, evaluates individual generated settlements against human-defined criteria. We adopt memorization checks as a disqualification gate; human judgment of exterior views faces the same observation limit as the metrics we audit, which is why our positive control changes the view (cutaways) rather than the judge. Vision-language judges of 3D generation [32] also read rendered views, and we score one as a third exterior observer.

3. Background and Problem Setting

A *structure* here is a map $x : \{1, \dots, d\}^3 \to \mathcal{V}$ from a voxel grid to a categorical block vocabulary $\mathcal{V}$, with one symbol reserved for air. An *arm* is a finite set of structures with a name and a provenance record: a generative model's samples, a procedural generator's output, or real data under a known corruption. Evaluation compares an arm's empirical distribution Q against a fixed reference P of real structures. Two features of this setting differ from image generation. Sample sizes are in the hundreds rather than the tens of thousands, and arms may differ in size, so a statistic whose null depends on n ranks by sample size.

Both metrics we report are kernel two-sample distances. For a feature map ϕ and a kernel k,

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}\, k(\phi(x), \phi(x')) + \mathbb{E}\, k(\phi(y), \phi(y')) - 2\,\mathbb{E}\, k(\phi(x), \phi(y)), \quad (1)$$

with $x, x' \sim P$ and $y, y' \sim Q$ [12]. The unbiased estimator of Eq. 1 omits the diagonal terms $k(\phi(x_i), \phi(x_i))$ from the within-set means. The biased estimator keeps them, so its null on two disjoint real halves decays with n. KID [4] is the unbiased estimator with a polynomial kernel on encoder features, and FID [14] fits a Gaussian to each feature set instead. A *render-and-embed* metric is any of these with $\phi = (\text{encoder}) \circ (\text{render})$, so the structure reaches the statistic only as a set of images.

Raw distances across features can differ by orders of magnitude, so we report all distances in *noise floors*. For each feature and sample size n, the noise floor is defined as the standard deviation of the distance obtained from repeated real-vs-reference draws under the same experimental run. Thus, a value of 1.0 indicates that the observed distance is one standard deviation of the metric's sampling variability. Because the sampling variability changes with n, the noise-floor unit is n-dependent (Supp. Sec. D.2). Comment: check all appendix refs

We treat a metric as *valid* for ranking when it has three properties.

1. **Ordering**: its value rises monotonically with the severity of a known corruption.
2. **Invariance**: its value does not move under a transformation that leaves quality unchanged.
3. **n-stability**: its null on real-vs-real draws does not drift with n.

Sensitivity, how *much* damage a metric resolves at a given n, is statistical power rather than measuring the right quantity, so we record it alongside the three rather than as a fourth.

4. Metric Validation

4.1. Probes

FILL-INTERIOR fills every enclosed air cell of a real build with that build's own dominant material, which keeps the silhouette bit-identical and adds only interior mass. Under a translucent renderer it does change pixel values inside that silhouette (§6.2) and can only be observed externally when a structure contains this transparency (e.g. a window). On 200 held-out real houses, it adds a median of **33**% blocks (mean 37%) and alters **94**% of these builds. The build's own material keeps the corruption from showing as a palette anomaly. The probe reaches only *sealed* air, air unreachable from the bounding box without passing through solid, so air reachable through a doorway survives; per build, the sealed share of a build's own interior volume has a median of 100% and a mean of **85.5**% (Supp. Sec. B).

HOLLOW-SHELL is the converse. It keeps a block only if one of its six faces borders air 6-connected to the outside, so it removes the solid behind the exterior shell where FILL-INTERIOR fills the sealed air, and it removes a median 13.4% of blocks from 98.4% of builds. Four further controls damage appearance or diversity rather than geometry. MATERIAL-SHUFFLE permutes the block multiset across a build's own occupied cells, preserving geometry exactly, and MONOCHROME repaints every block with one material. TRAIN-COPY (verbatim training builds) and SINGLE-MODE (one build repeated n times) damage nothing a distribution distance objects to.

4.2. MV- and V-KID

Both realism metrics are the unbiased estimator of Eq. 1 and differ only in ϕ (Supp. Fig. 1). **MV-KID** is our stand-in for the render-and-embed family with four orthographic views of each structure at 224px, embedded with DINOv2-B [26]. **V-KID** applies the same estimator to a 77-dimensional descriptor read directly off the voxel grid, examining the distribution over the 55 D4 orbits of $2 \times 2 \times 2$ occupancy patterns, a 5-bin wall-thickness histogram, an 8-bin normalized height profile, and 9 global scalars, in 15 to 18 s for several thousand builds on one CPU process (Supp. Sec. A, Supp. Sec. I.2). V-KID reads occupancy only, so under MATERIAL-SHUFFLE and MONOCHROME its value is *bit-identical* to the real row. It measures what the render tier cannot, and not structural quality outright.

4.3. Blocking Gates

Every candidate metric is scored on rungs of known damage, in units of its own noise floor. A metric that fails ordering, invariance or n-stability is reported as *gated*, with the failing gate named, and may not rank anything. Each gate restates a prior criterion, and its threshold predates V-KID (Supp. Sec. C.3). Applying this to the seven statistics of Supp. Tab. 2 and the biased MMD estimator settles three things (Supp. Tab. 1). First, we clearly see the render metric reads appearance rather than structure as MATERIAL-SHUFFLE preserves geometry exactly and is yet among the worst-scoring rungs, beaten only by MONOCHROME. Second, the biased MMD estimator is unusable at variable n: its null on two disjoint real halves reads 0.289 at n=16, 0.071 at n=64 and 0.014 at n=400, while the unbiased estimator reads 0.000 at every n. A leaderboard mixing arm sizes therefore ranks by sample size. Third, precision, density and recall fail monotonicity under block noise, and Fréchet distance and coverage fail n-stability. Only the unbiased KID and MMD-RBF clear each condition.

4.4. Aggregation and Disqualification

The gates decide which metrics may speak. Turning several that pass into one ranked number is a separate and more arbitrary problem. BlockScore is *one* instantiation: the **maximum** over five pillars, each divided by its own in-run scale, so there are no weights to tune. The property we defend is the maximum rather than the mean. The two tiers are blind in complementary directions (§6.4), so averaging lets an arm buy a catastrophic pillar with a good one, and the trade the render tier invites is the dangerous one: perfect surfaces over a solid interior. Supp. Sec. D.1 records two properties of the implementation: three of the five pillars (appearance, geometry, palette) are in true noise floors and two are in a coarser unit that never determines the maximum, and the null of a maximum is not zero.

Novelty and diversity are **disqualification gates** rather than weighted terms. An arm that trips one is refused a rank and shown with its reason, so a model cannot offset memorization with realism. The risk is concrete here: a pre-split autoregressive checkpoint from prior work, none of the arms scored here, duplicated roughly 31% of training builds at intersection over union (IoU) $\geq$ 0.95. Novelty is tested against the training split and, because a set can also repeat itself, within the set (Supp. Sec. L.1). Every arm must also declare the split it trained on, the requirement Supp. Fig. 11 turns on. Gates need auditing as much as metrics do (Supp. Sec. C.4).

5. Experimental Setup

Our test medium is Minecraft houses as they represent discrete 3D structures over a categorical block vocabulary, at a scale where a real corpus exists and interiors are semantically meaningful. We pool three public sources onto a shared vocabulary: a GrabCraft scrape, the 3D-Craft house traces [8, 11] and the text2mc dump. A quality gate and variant-aware deduplication leave **2,661** houses in a 32^3 box (Supp. Sec. E.1).

Public build corpora ship sibling structures, such as

"American Middle Class House 10 / 22 / 9", and a random row split scatters a family across train and validation. Over 20 seeds, a mean of **28.7%** of validation items share a build family with training (s.d. 2.7, range 23.1-35.8). Leakage at that rate inflates the held-out floor and deflates every memorization statistic at once. Our canonical split is group-aware over 2,054 families at 0% cross-split family overlap, pinned and released: train 1,863 / validation 399 / test 399. Comment: what is this next line? The calibration draw needs the same care, since taking the first n test builds rather than a shuffled sample inflates the measured noise floor by $\mathbf{10.9}\times$ (Supp. Sec. E.3).

The suite scores three kinds of arm against one fixed real reference (Table 1). *Probes* are real data with a known corruption (§4.1), *baselines* are five seeded generators and *submissions* are generative models. Baselines are fitted on the training split only, since one that had seen validation or test would hold an unearned advantage.

VOXEL-AR and VOXEL-AR-U are our own autoregressive transformers over merged-piece tokens, differing in whether builds are canonically oriented. Dream-Cubed [22] is a published 279M-parameter masked discrete diffusion model over 32^3 Minecraft chunks, run from its Apache-2.0 village checkpoint and trained on no source of our corpus. The protocol run also scores two LoRA fine-tunes of a 1.5B code model, LLM-FT and LLM-FT-GEO, as baselines for the case study of Supp. Sec. L.2; they are not on the leaderboard (Supp. Sec. L.2).

Every control is re-scored in every run rather than cached, because a distance means nothing until a floor has been measured under *this* run's corpus, split, n and reference size. The leaderboard scores every arm at n=256 and the probe, ladder and dose analyses run at n=128, all against the 399-build reference, and significance comes from an exact label-permutation test with Holm correction [15] over all arm pairs. The encoder ablation uses n=256 against a 512-build reference, with the noise floor taken from 48 fresh real draws. The second domain is 47,389 LEGO brick abstractions of ShapeNetCore objects on a 20^3 grid, 44% of which contain a sealed void, scored at n=256 against a 450-build reference. StableText2Brick and BrickGPT are MIT-licensed (Supp. Sec. M).

6. Results and Discussion

6.1. Every Observer Misses the Interior

Table 2 scores the two interior probes and one appearance control under every kind of observer the field uses: an image metric on exterior renders, learned and classical 3D metrics on the surface an exterior ray reaches, and a vision-language judge shown exterior views. None separates FILL-INTERIOR or HOLLOW-SHELL from real data, and every one that reads appearance separates MATERIAL-SHUFFLE. Each recovers the interior probes once its observation includes the interior, and V-KID, which reads the voxel grid, separates them by hundreds of noise floors.

A metric that is merely insensitive to interiors still moves when the interior changes. MV-KID does not. On the occupancy-noise and mixture axes it is monotone over 185 and 549 noise floors at n=128. On the interior axis its range end to end is under half a noise floor, 0.48 at n=64 and 0.34 at n=128, where V-KID reads 133 and 287 on the identical rungs of the same dose run (Supp. Sec. F). Raising n multiplies the geometric signal by 2.2 and leaves the render signal at its sampling floor. Both metrics rise on the mixture axis, a real arm progressively replaced by bad samples, which is how a generator actually fails.

The flatness is not a property of the one statistic we report. Supp. Fig. 4 places both metrics on eight rungs of known damage in one run, and Supp. Tab. 2 puts seven render-space statistics on one LEGO ladder. Six of them place the deletion of a fifth of a structure at 7–14 noise floors and one-minus-recall places it at 1.4, while none of them places filling the interior above 2.1. The blindness belongs to the representation they read rather than to any one of them.

Leaving the image does not by itself escape the null space. We score Uni3D, ULIP-2, chamfer and EMD on point clouds sampled three ways, on our houses and on 290 multi-room Objaverse buildings (Supp. Sec. I). Sampled from the faces an exterior ray reaches, every point-cloud metric is blind to interior fill and removal, within 1.6 noise floors in every cell at both seeds; sampled over all faces, the published default, Uni3D places a filled Minecraft interior at +21.7. Surface-fitted generators emit exterior shells, on which the two samplers coincide. A metric inherits the null space of what its sampler observes. Of fourteen structural metrics, including these encoders on interior-inclusive points and the voxel IoU and chamfer MMD of voxel generation, only V-KID passes every gate, as do generic orbit-averaged occupancy descriptors (Supp. Sec. J), and the voxel metrics rate the filled corpus closer to real than real data itself (Supp. Sec. I.2).

An exterior-view judge shares it too. A pre-registered GPT-4o pairwise judge prefers real over FILL-INTERIOR at chance on exterior views and at 0.70 once the roof is removed. It agrees with both metrics that PATCH-STITCH beats GABLED-HOUSE (0.72), yet ranks held-out real sixth of eleven arms, so it is a third exterior observer rather than ground truth (Supp. Sec. I.3).

6.2. It Is the Observation, Not the Encoder

The finding could indict DINOv2 at four views rather than render-and-embed as a class. It does not (Figure 2, Supp. Tab. 10). Four other encoders, CLIP ViT-B/16

Table 1. The arms, used by these names throughout; recipes in Supp. Tab. 8.

probes (real data, corrupted)		**baselines** (procedural)	
REAL	the floor: held-out real	RANDOM-VOXELS	random voxels at real density
FILL-INTERIOR	every interior cell filled	HOLLOW-BOX	hollow box, walls and roof
HOLLOW-SHELL	interior blocks removed	GABLED-HOUSE	box, pitched roof, door, windows
MATERIAL-SHUFFLE	blocks permuted, shape kept	PATCH-STITCH	8^3 patches, different builds
MONOCHROME	one material, shape kept	TRAIN-COPY+NOISE	a training build, 10% displaced
TRAIN-COPY	verbatim training copies	VOXEL-AR	a native token-based model
SINGLE-MODE	one build repeated n times	Dream-Cubed	a published diffusion model

Table 2. **Every observer misses the interior.** Separation from held-out real Minecraft houses in each run's own noise floors; GPT-4o rows give the probability that the judge prefers the real build (0.5 is chance). *vis* samples the faces an exterior ray reaches first and *cut* the faces the cutaway views show. MV-KID and V-KID: n=256 leaderboard, HOLLOW-SHELL scored against its reference and noise floor; point clouds: n=128, seed 0 (Supp. Tab. 12); cutaways: Supp. Tab. 10; judge: Supp. Tab. 14.

observer	FILL-INTERIOR	HOLLOW-SHELL	MATERIAL-SHUFFLE
exterior observation			
MV-KID, 4 views	**+0.35**	**+0.04**	+422.43
Uni3D KID, *vis*	**−0.1**	**+0.1**	+19.0
ULIP-2 KID, *vis*	**−0.1**	**−0.1**	+30.6
chamfer MMD, *vis*	**−0.1**	**−0.4**	−0.2
EMD MMD, *vis*	**+0.0**	**+0.0**	−0.0
GPT-4o, P(real)	**0.53**	**0.52**	—
observation that includes the interior			
MV-KID, floorplan cuts	+55.8	—	—
Uni3D KID, *cut*	+37.0	+13.5	+19.1
GPT-4o, roof off, P(real)	0.70	0.73	—
V-KID, voxel grid	+444.36	+265.4	+0.00

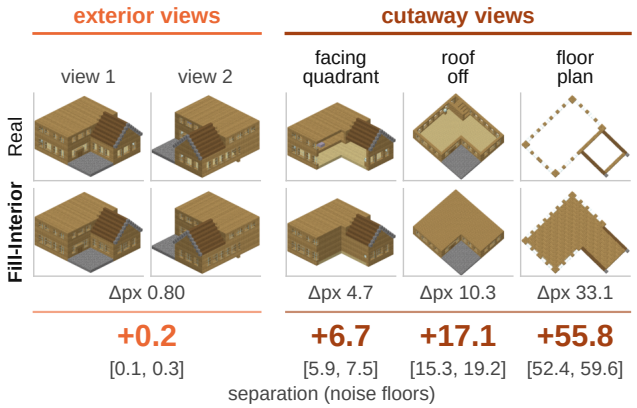


Figure 2. **The same encoder and statistic, only the views changed.** Each pair is a held-out build over its FILL-INTERIOR twin. *Left:* two of the four exterior views the standard protocol renders, visually indistinguishable for this build at a mean pixel difference of 0.8/255. *Right:* three cutaway protocols on the same builds, each with its separation in noise floors, with a paired-subsample 95% interval. DINOv2-B, unbiased KID, n=256 against 512 reference builds; full ablation in Supp. Tab. 10.

and B/32, ImageNet-supervised ResNet-50 and the Inception network FID itself uses, land between 0.0 and 0.2 noise floors, as does our DINOv2-B. Eight and sixteen views covering roof and underside read +0.2 and +0.1, perspective projection reads +0.1, and face-on elevations, which look straight through windows, read +4.4 at n=256 (Supp. Sec. G.2).

The cutaway rows are the positive control. The same encoder and the same statistic recover the interior at 6.7 to 55.8 noise floors once the view protocol stops hiding it, which makes this a statement about *information* rather than about features: **the standard view protocol does not put interior structure into the image, so no encoder downstream of it can score that structure.** A cutaway is not itself a good metric, because a section view discards the facade the render tier exists to read.

Under an *opaque* renderer interior damage provably never reaches the image: filling sealed air cannot alter any exterior view, so no encoder and no statistic can separate it at any sample size. Rendered with glass opaque, one fixed texture atlas and a CPU rasterizer, all 1,596 test-split views are bit-identical under both FILL-INTERIOR and HOLLOW-SHELL, so their MV-KID separation from real is exactly 0; Supp. Sec. H gives the proof.

Our default renderer is not opaque, and under it FILL-INTERIOR alters **91**% of the 1,596 test-split views, touching a median 1.23% of pixels at a median maximum channel difference of **128/255**. Doorways never leak, as air a doorway reaches is not sealed; glass in front of a filled void does, as does an atlas artifact of our mesh builder, and with the atlas fixed the fill shows in 67% of houses (Supp. Sec. H). Binned by how much of its silhouette the fill changes, each house's DINOv2 feature moves in proportion (Spearman 0.96), yet no bin moves MV-KID by more than 1.4 noise floors: the leak reaches single images, not the set statistic.

6.3. Both Directions and Three Domains

The corruption this field actually produces is the converse of FILL-INTERIOR: the hollow shells that surface-fitted mesh and implicit generators emit. It is equally invisible. Against

Table 3. **LEGO.** Separation from held-out real in each run's own noise floors.

arm	MV-KID	V-KID
ShapeNet bricks with a sealed void, n=256		
FILL-INTERIOR	$\mathbf{-0.01}$	$\mathbf{+118.6}$
MATERIAL-SHUFFLE	$+140.0$	$+0.0$
MONOCHROME	$+685.8$	$+0.0$
StableText2Brick test split, n=253		
FILL-INTERIOR	-0.09	$+36.4$
BrickGPT [28]	$+11.8$	$+51.5$

the leaderboard's reference and noise floor at n=256, MV-KID places HOLLOW-SHELL at $+0.04$ noise floors and FILL-INTERIOR at $+0.35$, while V-KID places the same two arms at $+265.4$ and $+444.36$ (Supp. Sec. G.2). The render metric is blind along the whole interior axis rather than to one hand-built probe.

The pattern is not a property of Minecraft houses. On the LEGO corpus (Table 3, Supp. Fig. 10), MV-KID places the solidified arm *below* the real floor and V-KID at $+\mathbf{118.6}$, while the render metric separates MATERIAL-SHUFFLE at $+140.0$ and MONOCHROME at $+685.8$. A published LEGO generator, BrickGPT [28], prompted with captions of 253 held-out test objects, separates on both tiers, $+11.8$ and $+51.5$, beside a solidified control at -0.09 and $+36.4$; its own validity and stability reproduce (Supp. Sec. K).

On 290 multi-room Objaverse buildings, the third domain, a DINOv2 kernel distance on four exterior views reads $+0.0$ for both removing the hidden faces and filling the hidden volume, at both seeds, while four cutaway views read 9.0 to 13.7 and V-KID 23.7 to 118.4 (Supp. Tab. 12).

6.4. Complementary Blindness and the Leaderboard

Table 4 shows the complementarity directly. FILL-INTERIOR reads $+0.35$ on appearance against $+444.36$ on geometry, and MATERIAL-SHUFFLE is the converse, $+422.43$ on appearance against *exactly* $+0.00$ on geometry. Noise floors grow with n, so we compare tiers by effect size, $\sqrt{\text{MMD}^2}$ as a fraction of RANDOM-VOXELS's under the same metric: FILL-INTERIOR sits at 0.03 on appearance, inside the 0.03 that fresh real draws reach, and at 0.78 on geometry (Supp. Sec. D.5). Each tier is fooled by the corruption aimed at it and rescued by the other, which is why BlockScore takes the worst pillar. The two failures differ in severity: the best attack we have on V-KID, PATCH-STITCH, buys near-perfect local occupancy statistics with globally incoherent structure and still sits at an effect size of 0.71 on geometry. The two feature spaces dissociate build by build as well: asked for each probe's nearest held-out real build, the render features return a filled house's own source 100% of the time and V-KID's features 26%, and for MATERIAL-SHUFFLE the proportions reverse, 26% against 100% (Supp. Fig. 8).

Dream-Cubed [22], the one published generator we score, is disqualified (Supp. Sec. L.1). Scored on the largest constructed component of each terrain chunk its village checkpoint emits, its builds copy nothing from our training split (duplicate rate 0.004) but copy each other: 59.8% have a near-copy in the same set (occupancy IoU ≥ 0.8 under rotation, mirroring and translation), against 0.8% for held-out real, and they are vanilla village pieces, 36% of them roofless enclosures. Neither novelty gate saw this, so we added a within-set near-twin gate under the duplicate gate's rule. It disqualifies Dream-Cubed at $+29.5$ spreads, the two parametric baselines, and the game's own village templates, which would otherwise score 202.95, ahead of every model we trained.

Deleting V-KID's size term and sealed-interior scalar leaves 197.3 of its 201.8 floors on FILL-INTERIOR in the protocol run (Supp. Sec. J.1). Every leave-one-family-out variant, every random half of its dimensions and descriptors with no hand design (random 3D CNNs, raw 3^3 patterns) pass all gates, the last at 257 to 437 floors on FILL-INTERIOR (Supp. Sec. J).

The sharper test is a control matched on the quantity in question. GROW-SHELL dilates the solid outward until it has added *exactly* the number of blocks FILL-INTERIOR adds to that build, which it achieves to within one block on 128 of 128 builds at a median of 196. Same builds, same material, same mass, and the only difference is whether the new mass goes inside or outside. The two metrics then part company completely. In that run MV-KID places GROW-SHELL at $+\mathbf{102.3}$ noise floors and FILL-INTERIOR at $+\mathbf{0.3}$, while V-KID places them at 214.5 and 201.8. No block count could produce that pair of rows.

The † row is our own autoregressive checkpoint, which had trained on 346 of the 399 test builds (87%). No distribution statistic and no memorization gate saw it, at a duplicate rate of 0.000; a required split declaration did, and retrained on the clean split the arm loses its margin over the language-model baselines (Supp. Sec. L.2). Benchmarks in this domain should require that declaration and refuse to rank arms that cannot make it. The maximum rule also names a different failure for each learned arm: the clean VOXEL-AR is geometry-limited, $+285.22$ against $+157.40$ on appearance, and PATCH-STITCH is limited by appearance, so a mean over pillars would hide both.

6.5. Limitations

The invisibility is a theorem about opaque rendering (Supp. Sec. H); for translucent renderers it is a measured attenuation, here on one renderer, and a path tracer sharing no code with ours is the natural replication. V-KID

Table 4. **The leaderboard** (n=256 per arm, 399 reference builds). Appearance and geometry are null-standardized z-scores: the arm's distance from held-out real over the standard deviation of 48 real-vs-reference draws at this n. That deviation shrinks by a factor of 0.49 to 0.72 when n doubles, so the z-scores are **not comparable across** n; the raw MMD^2 columns ($\times 10^3$) are stable in n. BlockScore is the **maximum** across five pillars; palette and the two structure pillars never determine it here, and the structure pillars use a coarser in-run scale (Supp. Sec. D.1). Rank is the 95% interval over 1,000 subsample replicates; an interval spanning several places marks a tie. † marks the contaminated checkpoint of Supp. Fig. 11, unrankable and shown for comparison. "Twins" is the within-set near-twin rate, the share of builds with a near-copy elsewhere in the same set (0.8% for held-out real); its gate disqualifies four rows, whose pillars are shown for comparison.

				z (noise floors)		raw MMD^2	
rank	arm	BlockScore ↓	worst pillar	appearance	geometry	MV-KID	V-KID
1	REAL *(anchor)*	0.00	—	+0.00	+0.00	0.001	−0.14
submissions							
(2)†	VOXEL-AR *(contaminated)*	229.29	geometry	+128.87	+229.29	0.208	12.08
3	VOXEL-AR *(clean)*	285.22	geometry	+157.40	+285.22	0.254	15.06
4	VOXEL-AR-U *(clean)*	345.10	geometry	+182.86	+345.10	0.295	18.25
5–7	PATCH-STITCH	431.70	appearance	+431.70	+368.21	0.696	19.48
10	RANDOM-VOXELS	1087.06	appearance	+1087.06	+727.48	1.752	38.63
probes and controls: known damage, not submissions							
5–7	**FILL-INTERIOR**	444.36	geometry	**+0.35**	**+444.36**	0.001	23.54
5–7	MATERIAL-SHUFFLE	422.43	appearance	+422.43	**+0.00**	0.681	−0.14
8	MONOCHROME	535.36	appearance	+535.36	+0.00	0.863	−0.14
9	TRAIN-COPY+NOISE	619.13	geometry	+271.94	+619.13	0.439	32.86
disqualified: refused a rank, with the reason							
	Dream-Cubed [22]	DQ	twins 59.8%	+165.71	+146.64	0.268	7.67
	vanilla village templates (n=174)	DQ	twins 67.2%	+110.21	+202.95	0.178	10.67
	GABLED-HOUSE	DQ	twins 20.7%	+535.62	+620.98	0.863	32.96
	HOLLOW-BOX	DQ	twins 24.6%	+864.00	+645.86	1.392	34.28
	SINGLE-MODE	DQ	mode collapse, +44.9 spreads on voxel diversity				
	TRAIN-COPY	DQ	memorization, +10.1 spreads on retrieval percentile				
	Dream-Cubed *(whole chunk)*	DQ	mode collapse, +24.0 spreads on voxel diversity: every chunk is a slab of terrain				

reads local geometry. Global coherence is a separate axis, which PATCH-STITCH, assembled from patches of real builds, exposes by scoring closer to real than the simple but coherent GABLED-HOUSE on both tiers, so it stays a standing control (Supp. Sec. J). V-KID is blind to materials by construction, and neither tier scores how an interior *looks*: cross-section views see it but rank arms by texture (Supp. Sec. G.2), so interior appearance is an open measurement. Detecting interior change is not ranking interior function: V-KID accepts cluttered interiors, and a walk-through test sides with MV-KID on 33 of 40 reversed generator pairs (Supp. Sec. L.4).

A noise floor is a run's own statistic and roughly halves when n doubles, so magnitudes compare within a run (n=256 for the leaderboard, 128 elsewhere) and only ranks and ratios carry across runs (Supp. Sec. D.2). The generator released with this corpus [8] reports a cumulative chamfer distance, which is exactly invariant to our probe (§2).

7. Conclusion

Evaluation for 3D structure generation was inherited from image generation, and the inheritance carries a specific defect. Rendering a structure and embedding the renders measures its surface, and a building is not its surface. Filling every interior in a corpus of real buildings is invisible to a multi-view kernel distance that resolves every corruption reaching the exterior. That invisibility survives every encoder, view budget and statistic we try, point-cloud metrics on visible surfaces and an image judge, and a cutaway view removes it: what the render metric misses is not the quantity of the change but its location.

A metric therefore earns the right to rank models by first resolving damage of known severity. The price is a handful of corruption arms scored beside the real reference in every run, and what it buys is that a result resting on a blind metric fails in the lab rather than in the literature. Structure that decides whether a generated object is usable, and that no camera can photograph, has to be measured directly, and the same audit is owed to mesh and implicit 3D generation.

LLM Usage Statement

We used a generative AI coding assistant substantially in preparing this work, and describe the division of labor rather than a blanket disclosure.

Where it was used. Drafting and editing the text of this paper; implementing the figure-generation code; implementing parts of the evaluation suite and the probes; and conducting literature searches, including the prior-art check that identified Bińkowski et al. [4]'s Appendix B.2 as the source of the mixture identity in Supp. Sec. D.5 and the limitation already stated by Merino et al. [22].

Where it was not. The research questions, the experimental design, the decision of what to measure, and the interpretation of results are the authors'. No experimental result, numerical value or table entry in this paper was produced by a language model: every number is emitted by the released code from a recorded run, and the figures read those runs directly rather than transcribing them.

Verification. Every citation was checked against arXiv or the official proceedings, and every numeral in the paper is re-derived from a recorded run by a registered check in the released code. The authors take full responsibility for the contents of the paper, including any remaining errors.

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